Big Data Analytics (BDA) - 5 Marks Solutions & Architectural Analysis

Autonomous Comprehensive Synthesis • Authored by Cognexa Autonomous AI

Subject:	Big Data Analytics (BDA) - 5 Marks Solutions & Architectural Analysis	**Date:**	Current Operational Cycle
Engine:	Cognexa Core 2.0 (LangGraph + Vector Memory)	**Security:**	Verified & Sandboxed

1. Executive Summary

Exhaustive academic examination solutions covering Big Data principles, the 5 V's framework, ecosystem tools, technological storage and processing challenges, data sources, and Lambda/Kappa distributed processing architectures.

Question 1: Summary of Big Data & The Five V's Framework (5 Marks)

Big Data refers to massive, high-velocity, polyglot data assets that exceed the processing capacity of traditional relational database management systems (RDBMS), requiring distributed storage and horizontal parallel compute clusters.

- 5 Core Points of Big Data: 1. Scale-Out Architecture: Relies on clusters of commodity hardware rather than expensive vertical mainframes. 2. Polyglot Schemas: Handles structured, semi-structured (JSON, XML), and unstructured (video, audio) streams. 3. In-Memory Computing: Modern engines (Apache Spark) leverage distributed RAM to accelerate compute 100x over legacy MapReduce. 4. Fault-Tolerant Distributed Storage: Automatically replicates data blocks (HDFS 3x default) across server racks. 5. Actionable Predictive Value: Drives automated machine learning and real-time operational decisions.
- The Five V's of Big Data: 1. Volume: Massive scale measured in Terabytes, Petabytes, and Exabytes. 2. Velocity: Extreme speed of data generation and required sub-second stream processing. 3. Variety: Diversity of formats (relational tables, logs, sensor signals, graph relationships). 4. Veracity: Data quality, trustworthiness, and noise reduction across noisy telemetry. 5. Value: The ultimate return on investment (ROI) derived by transforming raw telemetry into business intelligence.

Question 2: Key Technological Challenges in Managing Big Data (5 Marks)

Operating distributed Big Data systems involves severe architectural trade-offs across storage, networking, compute synchronization, and enterprise governance.

- Challenge 1 - Scalability Ceilings & Network Bottlenecks: Moving petabytes of data across distributed switch fabrics creates network saturation. Mitigated by data locality (bringing compute to where data blocks are stored).
- Challenge 2 - Schema Evolution & Dynamic Variety: Continuous updates to mobile app telemetry, IoT sensor firmware, and third-party APIs cause schema drift. Requires schema-on-read abstractions and schema registries (Apache Avro, Protobuf).
- Challenge 3 - Distributed Consistency & CAP Theorem: Reconciling data across geo-distributed nodes requires trade-offs between Consistency, Availability, and Partition Tolerance. Systems must adopt Eventual Consistency models for high availability.
- Challenge 4 - Data Privacy, Security & Compliance: Managing sensitive personal identifiable information (PII) at petabyte scale requires automated column-level encryption, role-based masking, and strict compliance with GDPR, HIPAA, and CCPA regulations.

Question 3: The Big Data Ecosystem Architecture (5 Marks)

The Big Data ecosystem comprises specialized, interoperable open-source frameworks engineered across ingestion, distributed storage, batch/stream processing, and analytical query execution.

- Data Ingestion Tier: High-throughput distributed event streaming via Apache Kafka, log aggregation with Apache Flume, and bulk relational database extraction with Apache Sqoop.
- Distributed Storage Tier: Hadoop Distributed File System (HDFS) splitting files into 128MB blocks across DataNodes, cloud object stores (Amazon S3, Google Cloud Storage), and NoSQL stores (Apache HBase, Cassandra).
- Processing & Compute Tier: Apache Spark for in-memory resilient distributed datasets (RDDs), Apache Flink for true low-latency event-time streaming, and Hadoop MapReduce for legacy long-running batch jobs.
- Data Lakehouse & Query Tier: ACID transactional metadata layers (Apache Iceberg, Delta Lake) combined with interactive massively parallel processing (MPP) query engines (Trino, Presto, Apache Hive).

Question 4: Architectural Characteristics of Big Data Systems (5 Marks)

Big Data systems exhibit unique structural traits that differentiate them from classical transactional database architectures.

- Horizontal Scalability (Scale-Out): Seamless addition of compute and storage nodes without system downtime, delivering linear performance scaling.
- Fault Tolerance & Self-Healing: Automatic node heartbeats and block replication; if a DataNode crashes, the NameNode automatically orchestrates re-replication of lost blocks to healthy nodes.
- High Concurrency & Massively Parallel Processing (MPP): Workloads are partitioned into thousands of concurrent tasks executed in parallel across worker nodes.
- Decoupled Compute and Storage: Modern cloud architectures separate storage (S3/GCS) from compute (ephemeral Spark/Kubernetes clusters), maximizing resource efficiency and minimizing infrastructure costs.

Question 5: Three Major Sources of Big Data (5 Marks)

Big Data originates from three primary operational paradigms across modern digital society.

- Source 1 - Social & Human-Generated Data: User interactions on social media platforms, search engine queries, video streaming platforms (YouTube, TikTok), emails, blog posts, and audio recordings. Characterized by high variety and unstructured natural language.
- Source 2 - Machine-Generated & IoT Sensor Telemetry: Continuous data streams emitted by industrial IoT sensors, GPS navigation transponders, smart utility meters, RFID inventory scanners, automated server log files, and medical telemetry. Characterized by extreme velocity and continuous time-series patterns.
- Source 3 - Enterprise Transactional Data: High-volume digital transaction logs generated by banking payment gateways, credit card processing networks (Visa/Mastercard), Point-of-Sale (POS) retail checkouts, and enterprise airline reservation systems. Characterized by structured relational schemas and high integrity requirements.

Question 6: Big Data Processing Architecture (Lambda vs Kappa Architectures) (5 Marks)

Distributed processing architectures harmonize high-velocity real-time event streaming with massive historical batch accuracy.

- Lambda Architecture Overview: Dual-pipeline processing paradigm consisting of 3 layers: 1. Batch Layer (HDFS + Spark/Hadoop storing immutable master dataset and precomputing batch views). 2. Speed Layer (Storm/Flink processing recent data in real-time to compensate for batch latency). 3. Serving Layer (HBase/Cassandra merging batch and real-time views for low-latency queries).
- Limitations of Lambda Architecture: Demands maintaining and synchronizing two completely separate codebases (one for batch, one for streaming), increasing engineering complexity and bug surface.
- Kappa Architecture Evolution: Replaces the dual-path with a single, unified stream processing log (Apache Kafka + Apache Flink). Historical re-computations are handled simply by replaying the Kafka log from offset 0, eliminating dual-codebase maintenance.
- Modern Lakehouse Architecture: Modern enterprise standard combining cloud object storage (S3) with ACID transaction layers (Apache Iceberg) and unified compute engines (Apache Spark Structured Streaming).